R16

Code No: 135BY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
COMPUTER GRAPHICS
(Common to CE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Compare raster-scan systems with random scan systems used in computer graphics.
b) Explain about Bresenham's mid-point circle generating algorithm. [7+8]

2.a) Write DDA line drawing algorithm and explain with suitable example.
b) Describe about flood fill polygon filling algorithm. [8+7]

3.a) Summarize about different composite transforms used in 2-D geometrical transforms.
b) Explain the working of Sutherland –Hodgeman polygon clipping algorithm with suitable example. [7+8]

4.a) Describe about window to view-port coordinate transformation with an example.
b) Determine a sequence of basic transformations that are equivalent to the y-direction shearing matrix. [7+8]

5.a) Describe about 3-D objects representation using spline representation.
b) Compare Hermite curve, Bezier curve and B-spline curves. [8+7]

6.a) Describe about the following 3-D geometric transformations:
i) translations and ii) rotations
b) Summarize about sweep representations of 3-D objects. [8+7]

7.a) Explain about classification of different visible surface detection methods.
b) Describe about basic illumination models. [8+7]

8.a) What are the different computer animation languages used in computer graphics?
b) Explain about general computer animation functions with examples. [7+8]

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